# HOMEWORK 6

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**8.1.** True or false:

a) Every change of coordinate matrix is square;
b) Every chance of coordinate matrix is invertible;
c) The matrices $A$ and $B$ are called similar if $B = Q^T AQ$ for some matrix $Q$;
d) The matrices $A$ and $B$ are called similar if $B = Q^{-1}AQ$ for some matrix $Q$;
e) Similar matrices do not need to be square.

*Solution.* ■

**8.2.** Consider the system of vectors

$$(1, 2, 1, 1)^T, (0, 1, 3, 1)^T, (0, 3, 2, 0)^T, (0, 1, 0, 0)^T.$$

a) Prove that it is a basis in $\mathbb{F}^4$. Try to do minimal amount of computations.
b) Find the change of coordinate matrix that changes the coordinates in this basis to the standard coordinates in $\mathbb{F}^4$ (i.e to the coordinates in the standard basis $\mathbf{e}_1, \ldots, \mathbf{e}_4$).

*Proof.* ■

**8.3.** Find the change of coordinates matrix that changes the coordinates in the basis $1, 1+t$ in $\mathbb{P}_1$ to the coordinates in the basis $1 - t, 2t$.

*Solution.* ■

**8.4.** Let $T$ be the linear operator in $F^2$ defined by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x - 2y \end{pmatrix}$$

Find the matrix of $T$ in the standard basis and in the basis

$$(1, 1)^T, (1, 2)^T$$

*Solution.* ■

**8.6.** Are the matrices

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & 2 \\ 4 & 2 \end{pmatrix}$$

similar? Justify.

*Proof.* ■

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